Lesson Management & Scheduling System– Final Project

INFO361 - SYSTEMS ANALYSIS AND DESIGN

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1. Executive Summary

The Lesson Management & Scheduling System modernizes Halina's piano teaching operations by replacing manual scheduling, paper records, and inconsistent reminders with a centralized digital platform. The system includes lesson scheduling, automated reminders, payment tracking, a student database, and an intake portal for new student inquiries. All core deliverables, including requirements, diagrams, mockups, and analysis, have been completed. The remaining work consists of implementation and training. Future enhancements include mobile optimization, online payments, and expansion of bilingual UI.

2. Project Description

Halina is a private piano instructor managing 8–9 weekly lessons. Her current workflow relies heavily on paper-based scheduling, requiring her to text parents for reminders manually and track payments in notebooks. The proposed system introduces a digital lesson calendar, student database, automated reminders, payment logging, and an online inquiry form. The scope includes designing the system conceptually—not building it—while delivering requirements, feasibility, diagrams, interface prototypes, and evaluation.

3. System Request

The system request identifies the business need for a digital scheduling and lesson management platform to improve day-to-day operations, reduce missed lessons, reduce administrative burden, and maintain accurate payment histories. The request includes the business need, value, requirements, and constraints as documented in prior deliverables.

4. Business Case

The business case demonstrates clear value by reducing manual processes, improving communication, and providing Halina with a modernized, scalable system. The solution is cost-effective, using low-cost tools and a basic laptop. Intangible benefits include reduced stress, greater organization, and improved professionalism.

5. <u>Feasibility Analysis</u>

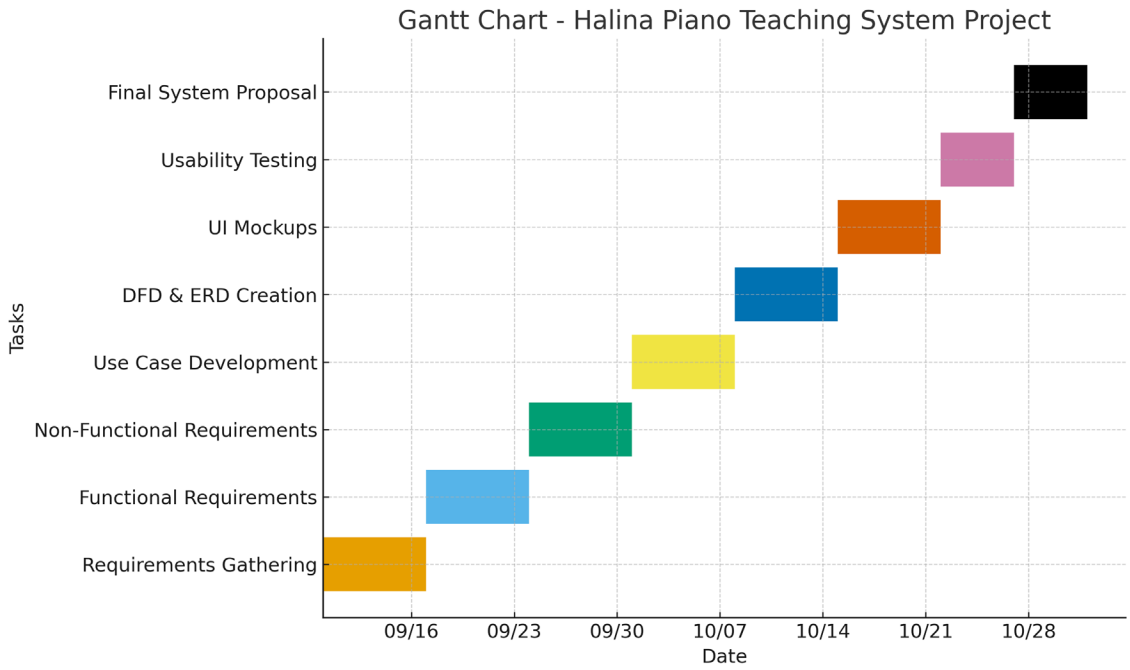
• Technical Feasibility: Simple tools (Google Calendar, Sheets, or custom apps) make the system low risk.

• Economic Feasibility: Estimated cost under $700 with high long-term value.

• Organizational Feasibility: Halina can be trained easily; bilingual support is needed, but manageable.

6. <u>Work Plan / Gantt Chart</u>

The work plan includes requirement gathering, analysis, design documentation, diagram creation, mockups, and testing. Tasks were completed on time following a structured workflow.



7. <u>Requirements Definition</u>

Functional and non-functional requirements documented in Assignment 5 and Requirements Report include:

- Lesson scheduling
- Student database
- Automated reminders
- Payment tracking
- Inquiry form
- Monthly reporting

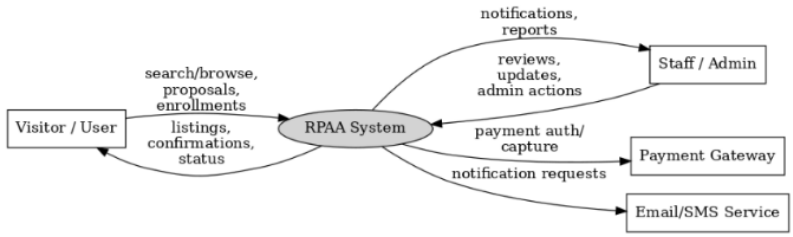
Non-functional requirements include usability, accessibility, performance, security, and bilingual options.

8. <u>Use Cases</u>

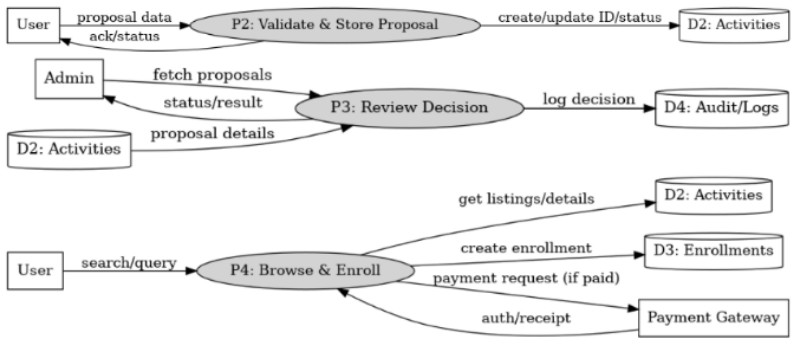
Use cases include Managing Lesson Schedules, Sending Automated Reminders, Recording Payments, Handling Intake Inquiries, and rescheduling or canceling lessons. Fully dressed and casual use cases have been documented and validated.

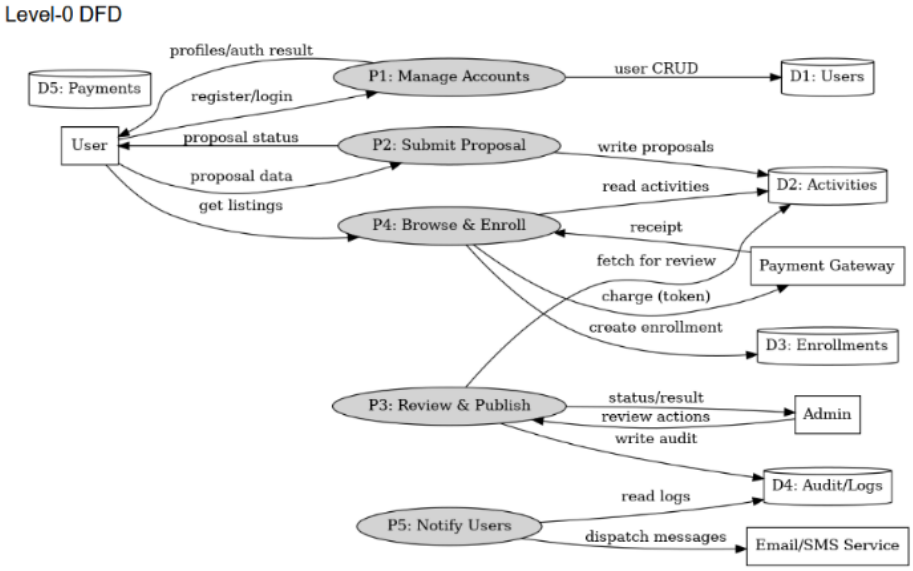
9. Data Flow Diagrams

Context Diagram

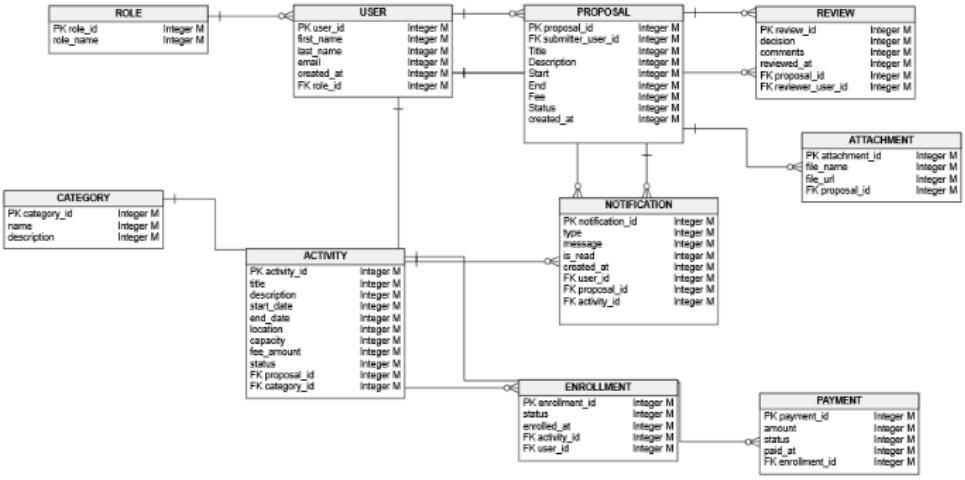


Data Flow Fragments (A, B, C)

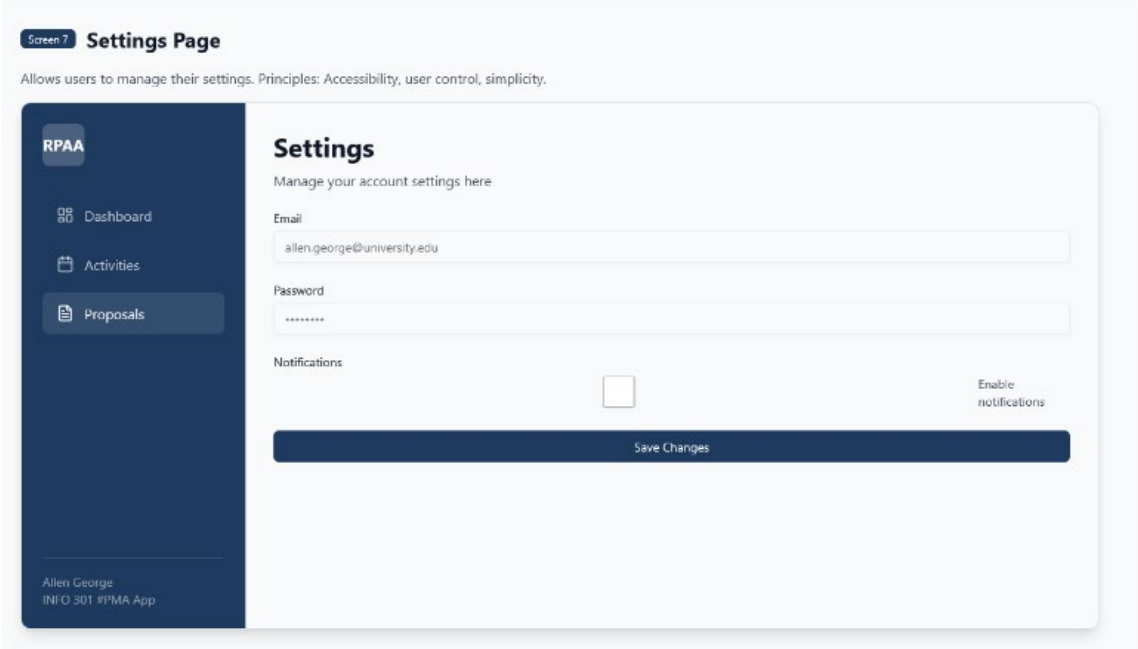


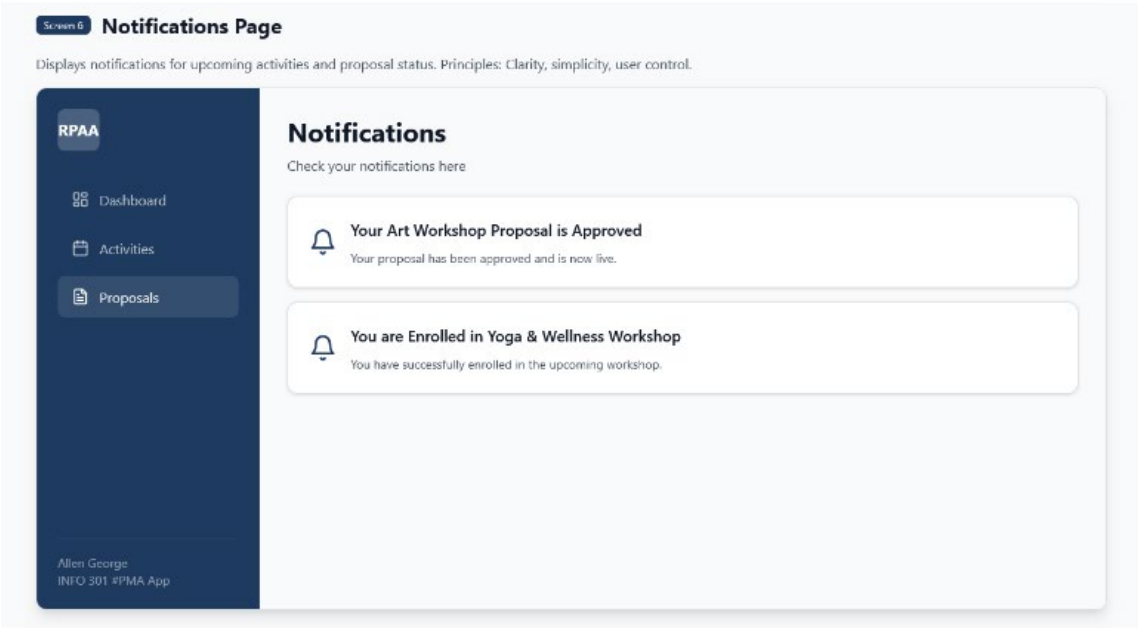


10. Entity Relationship Diagram



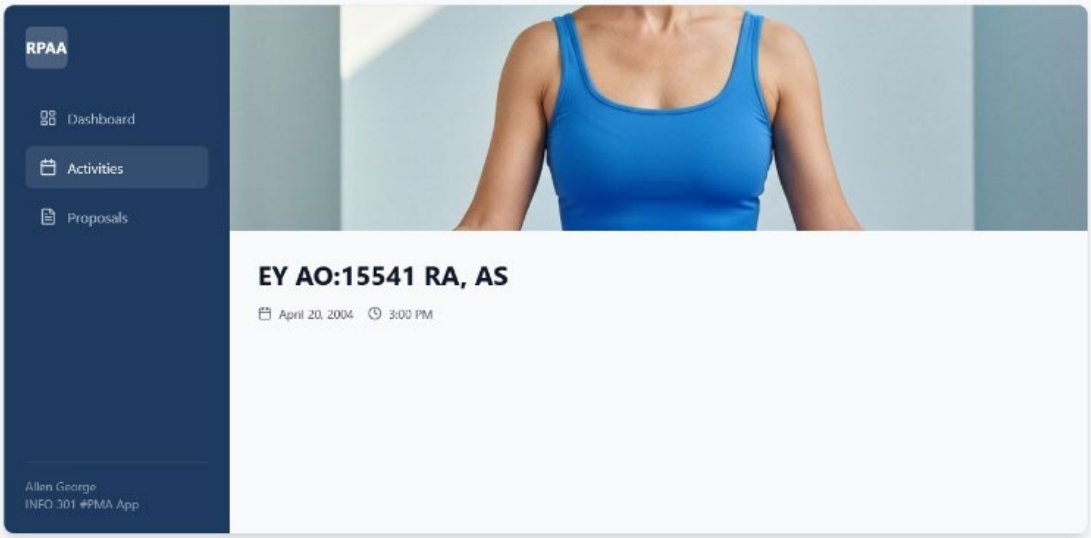
11. User Interface Prototypes





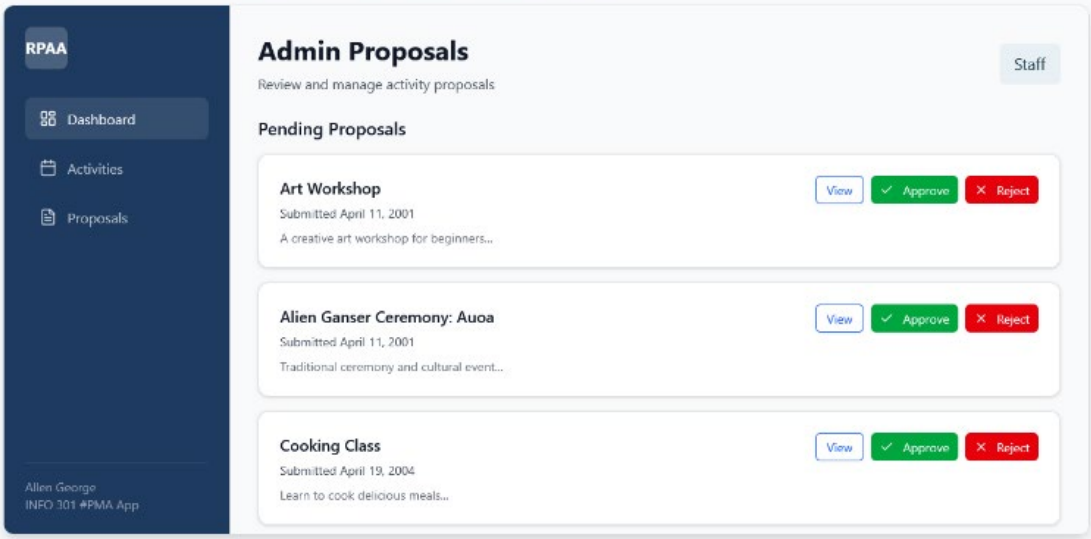
Screen 5 **Activity Details / Enrollment Page**

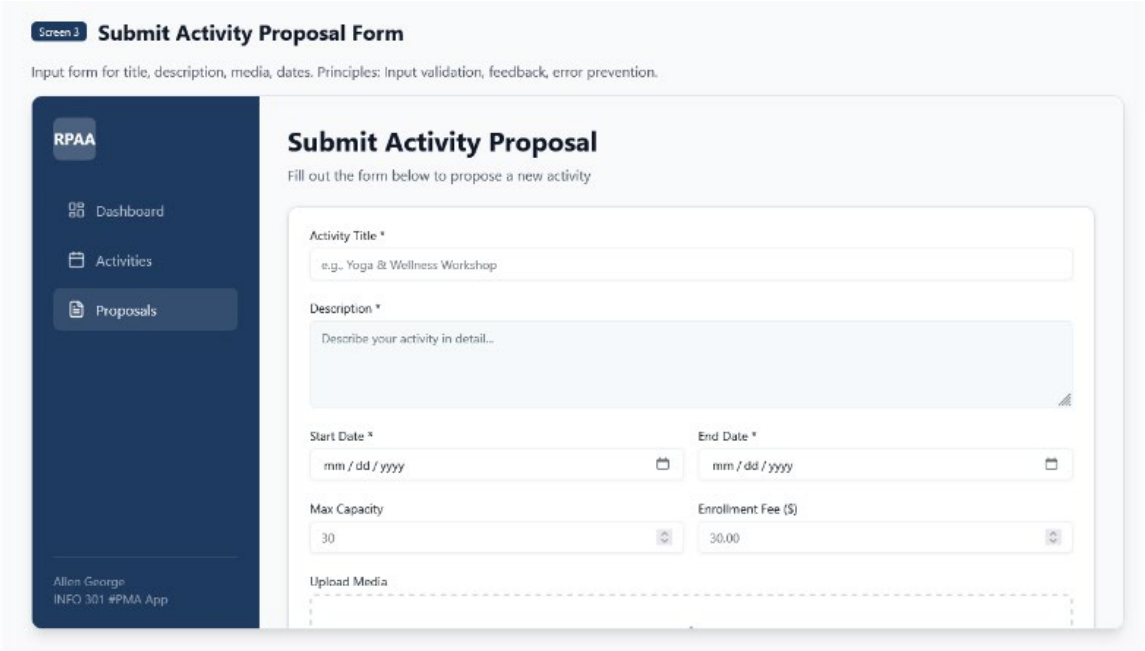
Shows published activity details and enrollment options. Principles: Consistency, easy navigation, feedback on actions.

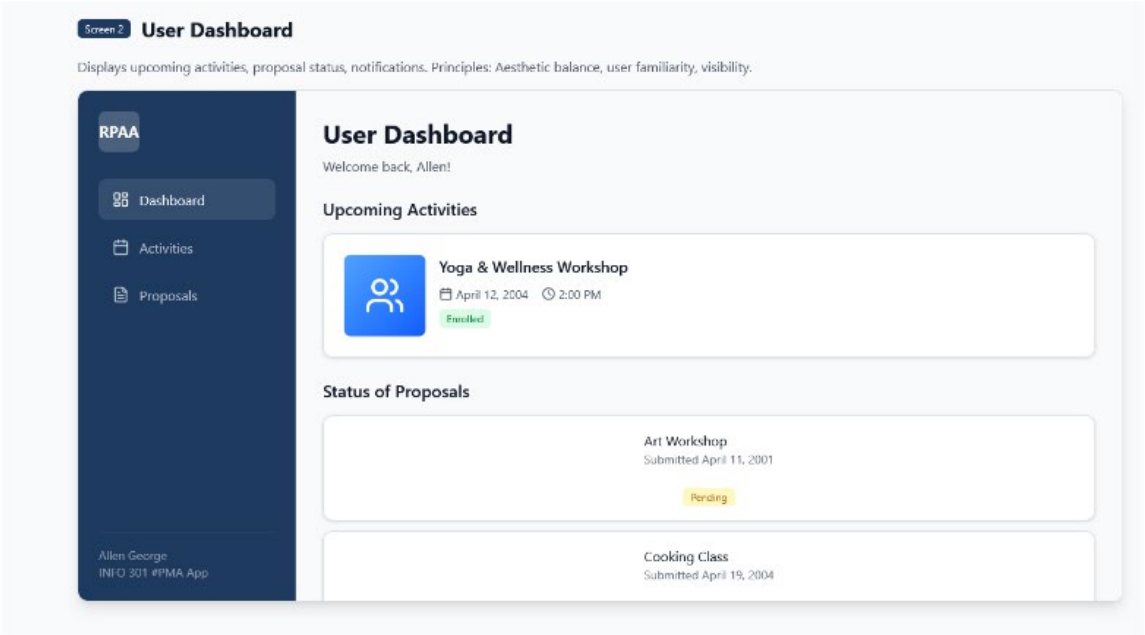


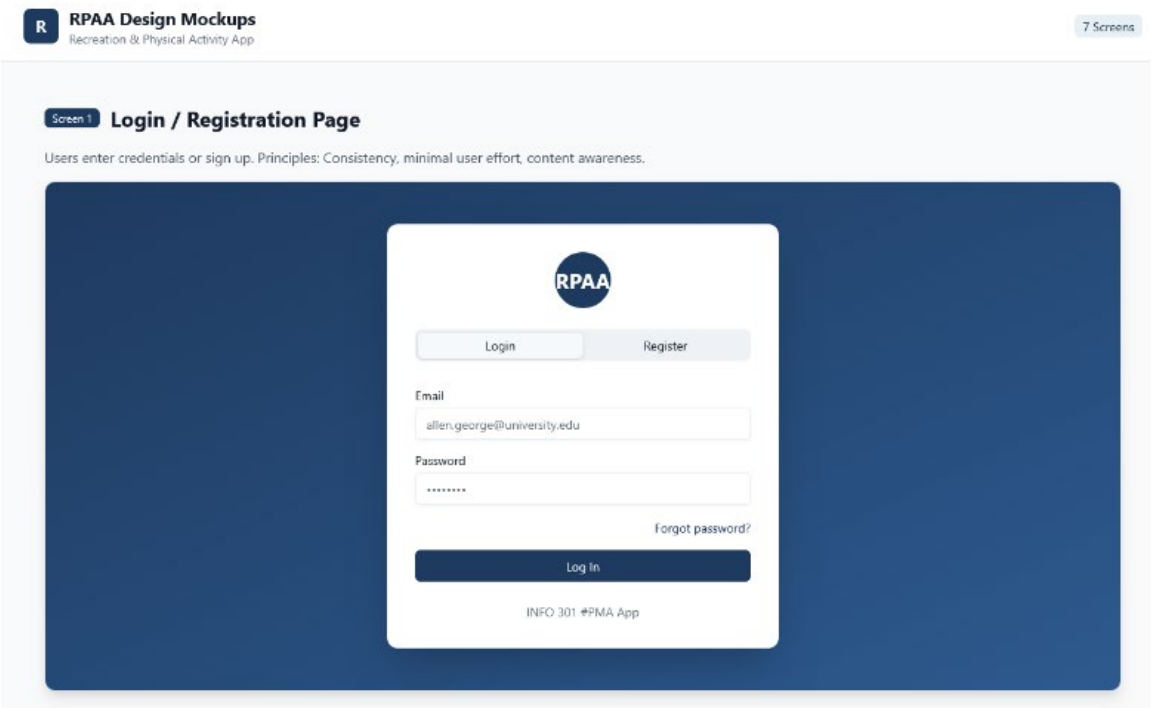
Screen 4 **Admin Review & Approval Panel**

Admin reviews proposals, approves/rejects. Principles: Content awareness, navigation clarity, error recovery.




Screen 3
Submit Activity Proposal Form
Input form for title, description, media, dates. Principles: Input validation, feedback, error prevention.
RPAA
Dashboard
Activities
Proposals
Allen George
INFO 301 #PMA App
Submit Activity Proposal
Fill out the form below to propose a new activity
Activity Title *
e.g., Yoga & Wellness Workshop
Description *
Describe your activity in detail...
Start Date *
mm / dd / yyyy
End Date *
mm / dd / yyyy
Max Capacity
30
Enrollment Fee ($)
30.00
Upload Media


Screen 2
User Dashboard
Displays upcoming activities, proposal status, notifications. Principles: Aesthetic balance, user familiarity, visibility.
RPAA
Dashboard
Activities
Proposals
Allen George
INFO 301 #PMA App
User Dashboard
Welcome back, Allen!
Upcoming Activities
Yoga & Wellness Workshop
April 12, 2004
2:00 PM
Enrolled
Status of Proposals
Art Workshop
Submitted April 11, 2001
Pending
Cooking Class
Submitted April 19, 2004



12. <u>Usability Test Report</u>

Two participants evaluated the UI mockups. Feedback included clearer button labeling, adding dashboard icons, and including confirmation popups. Adjustments were made accordingly.

13. <u>Recommendations</u>

The project is ready for implementation. Recommended enhancements include online payments, mobile application support, expanded reporting features, and full bilingual UI integration for English and Russian users.